C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 1: Properties and Trends **Paper 1**

#	Question	Answer
1	Appearance of F_2 at room temperature	Pale yellow gas
2	Appearance of Cl_2 at room temperature	Pale green gas
3	Appearance of Br_2 at room temperature	Red-brown liquid
4	Appearance of I_2 at room temperature	Grey-black solid
5	Colour of iodine vapour	Purple
6	Trend in boiling point down Group 7	Increases
7	Trend in electronegativity down Group 7	Decreases
8	Trend in oxidising ability of halogens down Group 7	Decreases
9	Trend in reducing ability of halide ions down Group 7	Increases
10	Why boiling point increases down Group 7	More electrons, stronger van der Waals

C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 2: Displacement Reactions **Paper 1**

#	Question	Answer
1	Strongest oxidising halogen	F_2
2	Strongest reducing halide ion	I^-
3	Colour of $Cl_2(aq)$	Pale green
4	Colour of $Br_2(aq)$	Orange
5	Colour of $I_2(aq)$	Brown
6	Colour of Br_2 in hexane	Orange
7	Colour of I_2 in hexane	Purple
8	Ionic equation for Cl_2 with Br^-	$Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$
9	Ionic equation for Cl_2 with I^-	$Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$
10	Ionic equation for Br_2 with I^-	$Br_2 + 2I^- \rightarrow 2Br^- + I_2$

C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 3: Silver Nitrate Test for Halides **Paper 1**

#	Question	Answer
1	Reagent to test for halide ions	Acidified $AgNO_3$
2	Acid used to acidify	Dilute HNO_3
3	Why the $AgNO_3$ is acidified	Removes CO_3^{2-} so Ag_2CO_3 cannot form
4	Why HCl is not used to acidify	Adds Cl^- ions
5	Colour of AgCl precipitate	White
6	Colour of AgBr precipitate	Cream
7	Colour of AgI precipitate	Yellow
8	Result with F^- ions	No precipitate
9	Silver halide that dissolves in dilute NH_3	AgCl
10	Silver halide insoluble even in conc NH_3	AgI

C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 4: Halides with Concentrated Sulfuric Acid **Paper 1**

#	Question	Answer
1	Gas formed from NaCl with conc H_2SO_4	HCl
2	Observation for NaCl with conc H_2SO_4	Steamy fumes
3	Gases formed from NaBr with conc H_2SO_4	HBr, Br_2, SO_2
4	Observation for NaBr with conc H_2SO_4	Steamy fumes and orange-brown fumes
5	Reduction product of H_2SO_4 with Br^-	SO_2
6	Reduction products of H_2SO_4 with I^-	SO_2, S, H_2S
7	Observation showing S forms	Yellow solid
8	Smell of H_2S	Rotten eggs
9	Observation showing I_2 forms	Purple fumes, black solid
10	Why Cl^- does not reduce H_2SO_4	Cl^- is too weak a reducing agent

C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 5: Uses of Chlorine **Paper 1**

#	Question	Answer
1	Equation for Cl_2 with water	$Cl_2 + H_2O \rightleftharpoons HCl + HClO$
2	Name of HClO	Chloric(I) acid
3	Type of reaction of Cl_2 with water	Disproportionation
4	Equation for Cl_2 with water in sunlight	$2Cl_2 + 2H_2O \rightarrow 4HCl + O_2$
5	Why chlorine is added to drinking water	Kills bacteria
6	Risk of chlorinating water	Forms toxic chlorinated hydrocarbons
7	Why chlorine is still used in water	Benefits outweigh risks
8	Equation for Cl_2 with cold dilute NaOH	$Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$
9	Name of NaClO	Sodium chlorate(I)
10	Use of NaClO solution	Bleach

C10 Group 7, the Halogens – Knowledge Organiser

AQA A-level Chemistry • Set 6: Equations **Paper 1**

#	Question	Answer
1	Equation for NaCl with conc H_2SO_4	$NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$
2	Equation for NaBr with conc H_2SO_4	$NaBr + H_2SO_4 \rightarrow NaHSO_4 + HBr$
3	Equation for HBr reducing H_2SO_4	$2HBr + H_2SO_4 \rightarrow Br_2 + SO_2 + 2H_2O$
4	Equation for HI reducing H_2SO_4 to SO_2	$2HI + H_2SO_4 \rightarrow I_2 + SO_2 + 2H_2O$
5	Equation for HI reducing H_2SO_4 to S	$6HI + H_2SO_4 \rightarrow 3I_2 + S + 4H_2O$
6	Equation for HI reducing H_2SO_4 to H_2S	$8HI + H_2SO_4 \rightarrow 4I_2 + H_2S + 4H_2O$
7	Half-equation for oxidation of Br^-	$2Br^- \rightarrow Br_2 + 2e^-$
8	Ionic equation for the silver chloride precipitate	$Ag^+ + Cl^- \rightarrow AgCl$
9	Equation for AgCl dissolving in NH_3	$AgCl + 2NH_3 \rightarrow [Ag(NH_3)_2]^+ + Cl^-$
10	Ionic equation for Cl_2 with cold dilute NaOH	$Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$